PRACTICAL / TUTORIAL 13

SELENIUM ENVIRONMENT SETUP

LAB PRACTICAL REPORT

Student Name	RUTVIK VALA
Enrollment Number	2504030400048
Program	BSc CS IT
Practical Topic	Selenium Environment Setup using SauceDemo

Includes workbook tables, setup/testing details, and available SauceDemo reference screenshots.

Implementation Details

Tutorial 13 – Selenium Environment Setup	
Create and configure Python, Selenium WebDriver, browser driver, IDE, and automation libraries.	
Parameter	Details
Project Name	Selenium Environment Setup
Application/Software	Sample web browser automation
Module Name	Environment verification and browser launch
SDLC Model	Agile
Programming Language	Python 3.x
IDE/Tool Used	VS Code or PyCharm, Google Chrome
Input/Test Data	https://www.google.com/ (or another permitted test URL)
Automation Tool	Selenium WebDriver
Python Version	Python 3.x (installed version)
Selenium Version	Selenium 4.x (pip show selenium)
WebDriver Used	Selenium Manager / ChromeDriver
Browser	Google Chrome
Test Framework	Pytest (optional)
Execution Date	Fill actual execution date

SDLC-STLC Mapping

SDLC–STLC Mapping Table			
Mapping software development phases to testing activities.			
SDLC Phase	Corresponding STLC Phase	Objective	Key Activities
Requirement Analysis	Requirement Analysis	Understand expected behavior	Identify browser launch and environment requirements
System Design	Test Planning	Plan testing approach	Choose Python, Selenium, browser and IDE
Development	Test Design	Prepare test scenarios	Write browser launch and title verification script
Testing	Test Execution	Verify environment and script	Run script and record browser/output results
Deployment	Test Closure	Document results	Save screenshots and execution report

Test Case Details

Test Case Detailed																		
Fill actual result and status after running the practical.																		
Field	Details																	
TC-01	REQ-01	Environment Setup	Python installation	Verify Python version	Environment	Black Box	High	Medium	Python installed	Terminal is available	Local computer	Windows / Chrome	python --version	Version is displayed	Fill after run	Not Run	Student	Actual date
TC-02	REQ-02	Browser Setup	Chrome browser launch	Open test URL using Selenium	System	Black Box	High	Medium	Python and Selenium installed	Browser closes after script	Local computer	Chrome	https://www.google.com/	Browser opens successfully	Fill after run	Not Run	Student	Actual date
TC-03	REQ-03	WebDriver	Automatic driver management	Confirm Selenium can start browser	Integration	Black Box	High	Medium	Internet access and Chrome installed	Driver session ends	Local computer	Chrome	webdriver.Chrome()	No driver setup error	Fill after run	Not Run	Student	Actual date
TC-04	REQ-04	Browser Verification	Page title verification	Verify page title is non-empty	System	Black Box	Medium	Low	Website is reachable	Browser is closed	Local computer	Chrome	Page title	Title is printed	Fill after run	Not Run	Student	Actual date

Test Case Execution

Test Case Execution				
Record the actual outcome for each step; do not mark Pass until verified.				
Step No.	Test Steps	Expected Result	Actual Result	Status
1	Open terminal	Terminal opens	Fill after run	Not Run
2	Run python --version	Python version is displayed	Fill after run	Not Run
3	Install Selenium using pip	Installation completes	Fill after run	Not Run
4	Run the Selenium Python script	Chrome browser launches	Fill after run	Not Run
5	Verify page title and close browser	Title is printed and browser closes	Fill after run	Not Run

Observation & Output

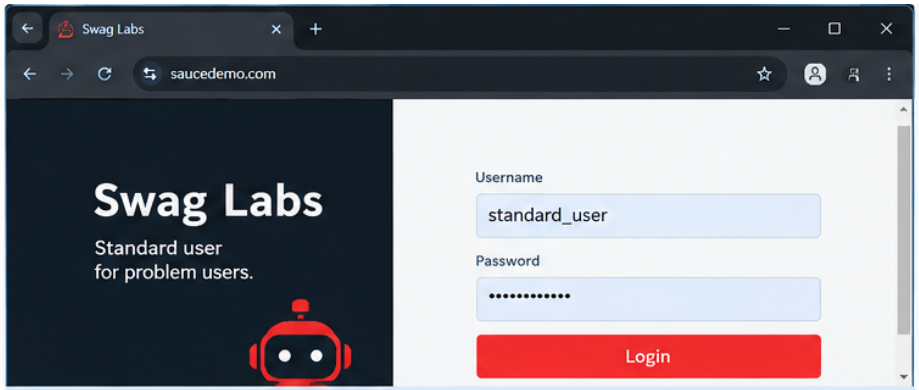
Observation and Output			
Attach actual screenshots after execution.			
Sr. No.	Observation	Expected Evidence	Screenshot / Remarks
1	Python environment check	Version shown in terminal	Paste/insert screenshot
2	Selenium installation	pip reports Selenium installed	Paste/insert screenshot
3	Browser launch	Chrome opens the selected test URL	Paste/insert screenshot
4	Page title verification	Page title is printed in terminal	Paste/insert screenshot
5	Execution completion	Script finishes without an unhandled error	Paste/insert screenshot

MCQ Answers

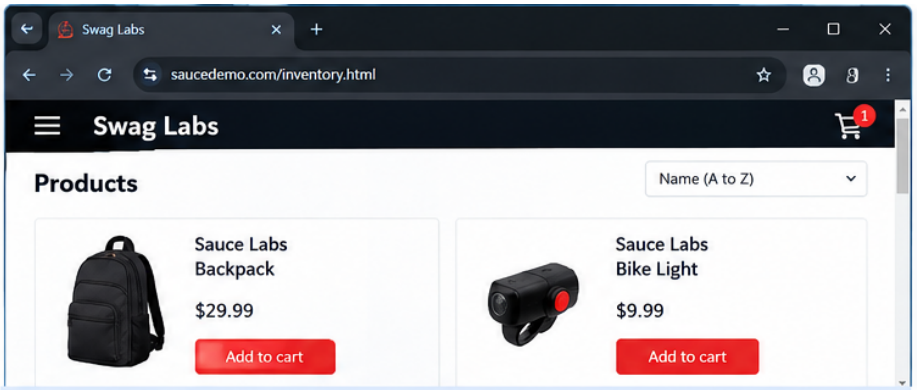
Questionnaire – Answers			
Tutorial 13 Selenium Environment Setup			
Q. No.	Question	Options	Correct Answer
1	Which library is used for browser automation in Python?	A) NumPy B) Pandas C) Selenium D) Matplotlib	C) Selenium
2	What is the purpose of ChromeDriver / WebDriver?	A) Database management B) Browser communication with Selenium C) Image editing D) File compression	B) Browser communication with Selenium
3	Which command installs Selenium?	A) install selenium B) pip install selenium C) python selenium D) setup selenium	B) pip install selenium
4	Which IDE can be used for Selenium script development?	A) MS Paint B) VS Code C) Calculator D) Notepad only	B) VS Code
5	What is the purpose of Selenium Manager?	A) Database management B) Driver download and configuration automation C) Browser removal D) Network monitoring	B) Driver download and configuration automation

SauceDemo Screenshot Evidence / Reference

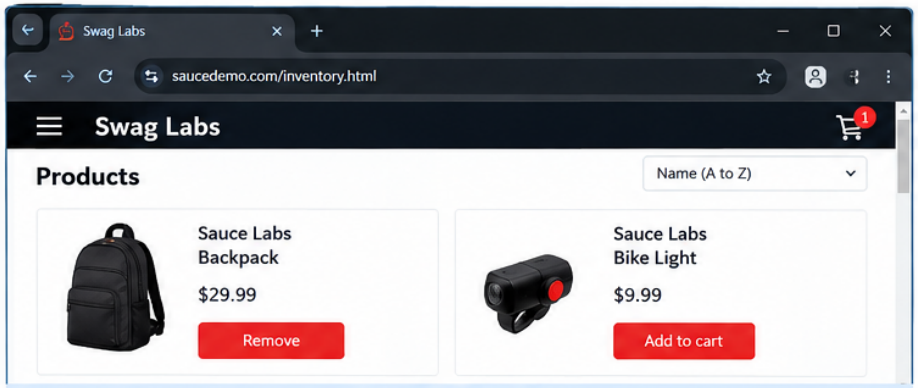
The image below is included as visual reference for browser navigation, login, product/cart interaction and console output. For exact Tutorial 13 environment-setup screenshots, replace/add screenshots captured from your own execution.



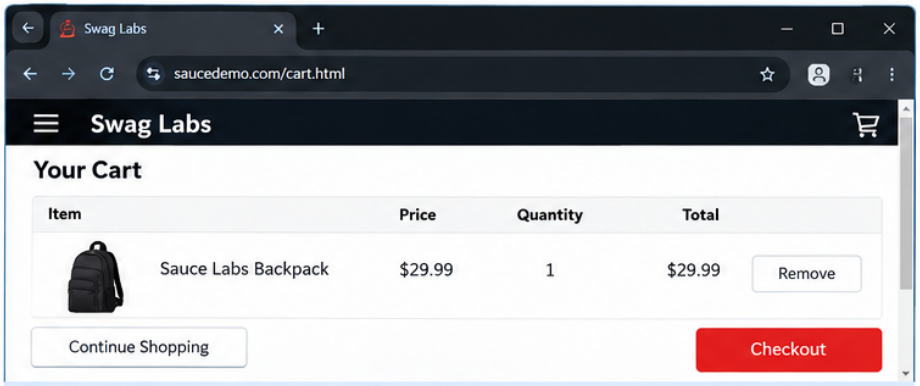
1. Browser Navigation – Login Page
SauceDemo website successfully opened.



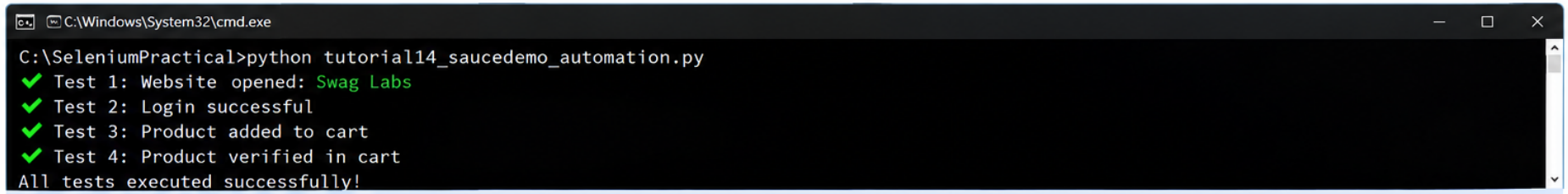
2. Login – Products Page
Valid credentials entered and login successful.



3. Web Element Interaction – Product Added
Sauce Labs Backpack added to cart.



4. Validation Testing – Cart Page
Product verified in cart successfully.



5. Test Execution – Console Output
All test cases executed and passed.

Conclusion

This practical documents the setup of the Selenium automation environment, including Python, Selenium WebDriver, browser configuration, and test planning. Run the practical locally and record actual outcomes before marking any test as passed. The included screenshot collage is reference evidence, not a claim that the current environment executed the tests.